

Week 9 Report Group 9 for DS340W

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Abstract—Real time win probability prediction has become increasingly valuable in sports analytics, yet existing systems provide limited interactivity and fail to incorporate comprehensive contextual features that influence game outcomes. This project develops a machine learning based sports analytics dashboard that provides continuously updated win probabilities during live NBA games while introducing a novel interactive what if simulation capability. Our system integrates multiple data sources including play-by-play event data, team Elo ratings, dynamic player statistics, and real time lineup compositions to capture both pregame team strength and in game momentum shifts. We employ ensemble learning techniques, combining logistic regression, gradient boosting, and neural networks, to predict win probabilities that update in real time as game events unfold. The dashboard’s interactive simulation module enables users to explore scenarios such as alternative substitution patterns, hypothetical scoring sequences, or foul situations and immediately observe their projected impact on win probability. Additionally, we integrate large language models to generate natural language explanations of probability shifts, making complex analytics accessible to both casual fans and professional analysts.

Index Terms—Sports analytics, real-time prediction, dashboard visualization, win probability, interactive simulation.

I. INTRODUCTION

THE purpose of this project is to create a data science solution that enhances the way live sports are experienced by both fans and analysts. Current trackers provide raw statistics and static win probability curves, but lack interactivity and predictive depth. For instance, ESPN’s Basketball Power Index (BPI) [2] offers pregame win probabilities but does not update dynamically based on in-game events or allow users to explore hypothetical scenarios. Similarly, FiveThirtyEight’s NBA predictions [3] focus primarily on season-long forecasting and pregame probabilities rather than real-time, play-by-play probability updates. While *Inpredictable* [6] developed a functional in-game win probability calculator for the NBA, it remains a passive tool that displays probabilities without enabling user interaction or scenario exploration. These existing systems, though valuable for providing baseline win expectations, do not offer the participatory element of simulating alternative game trajectories or explaining probability shifts in accessible terms, leaving a significant gap in how fans and analysts can engage with predictive analytics during live games. Our dashboard will fill this gap by offering continuously updated winning probabilities, presented through intuitive visualizations.

Beneficiaries of this system include:

- Fans, who will gain engaging, real-time insight into how likely their team is to win.
- Sports media and commentators, who can enhance live broadcasts with predictive analytics.

- Teams and analysts, who can benchmark in-game decisions against probability forecasts.

The novel aspect of our work is the interactive “what-if” simulator, which enables users to adjust game factors and immediately see projected changes in win probability. This adds a participatory layer that does not exist in existing win probability dashboards. In addition, we plan to leverage large language models (LLMs) to provide natural-language explanations of probability shifts and game scenarios, making analytics more accessible to fans and analysts.

A. Anticipated Technical Challenges

Several technical challenges are expected:

- 1) Real-time data ingestion from NBA APIs.
- 2) Synchronizing and cleaning play-by-play data streams.
- 3) Training predictive models that update probabilities in seconds.
- 4) Scaling the system to handle multiple games simultaneously.
- 5) Integrating the “what-if” module without disrupting live updates.

II. LITERATURE REVIEW

A. General Machine Learning Framework for Sports Prediction

Bunker and Thabtah [7] establish a comprehensive framework for applying machine learning to sport result prediction, with a particular emphasis on Artificial Neural Networks and their effectiveness in handling the non-linear relationships present in sports data. Their analysis identifies three critical data categories for successful predictive models: historical match outcomes, player performance metrics, and opposition characteristics. The authors highlight specific challenges in sports prediction that distinguish it from conventional classification tasks, including high dimensional feature spaces, class imbalance in outcome distributions, and the inherent dependencies between sequential game events. Their proposed framework addresses model evaluation through cross-validation strategies that account for sequential ordering, preventing data leakage from future information into training sets, a relevant consideration for live sports prediction systems.

B. NBA Win Probability Implementation

Inpredictable [6] presents a practical NBA win probability calculator utilizing locally weighted logistic regression, a non-parametric algorithm that assigns greater weight to training examples similar to the current game state. The model

employs four input variables: game time remaining, current point differential, possession status, and Vegas point spread. This feature selection demonstrates the tradeoff between bias and variance in sports prediction modeling, where a minimal feature set reduces model complexity and overfitting risk while capturing the most informative game state characteristics. The inclusion of Vegas point spread is particularly noteworthy as it incorporates the collective predictions of betting markets as a pregame team strength indicator. This approach aligns with the principle of using external signals to enhance model predictions, similar to how Elo ratings can be integrated as prior probabilities in sports prediction modeling.

C. Cross-Sport Win Probability Modeling

Robberechts [5] develops an in game win probability model for soccer using machine learning techniques that emphasize feature engineering and sequential modeling. His work addresses the challenge of capturing momentum and contextual game dynamics through time-series features and statistics that encode recent performance trends. The model uses gradient boosting methods, which are particularly effective for sports prediction due to their ability to model complex, nonlinear interactions between features while maintaining interpretability through feature importance metrics. This technique demonstrates the value of ensemble learning approaches that combine multiple weak learners to achieve robust predictions. Their framework also highlights the importance of probability calibration, ensuring that predicted probabilities accurately reflect true outcome frequencies.

III. DATASET AND DATA COLLECTION

A. Data Sources

Our system integrates multiple data sources: the NBA API Python library and FiveThirtyEight's historical Elo Ratings, to create a comprehensive feature set for win probability prediction. The data comes from multiple subpackages within the Python NBA API library. Datasets used include:

- **PlayByPlayV3** - A comprehensive dataset from the `nba_api` package containing detailed play-by-play logs for every game of the NBA season. This dataset serves as the foundation for predicting win probability in our models. Key features include unique game identifiers, time-stamped score updates, team information, and play descriptions that capture each significant event during the game.
- **ScoreboardV2** - Since many player and team statistical packages within the `nba_api` library require queries by either date or team identifier, data from this package was joined with the play-by-play dataset to enrich our feature set. This integration enables a seamless connection between game events and broader team and player statistics.
- **Elo Rating** - To quantify team strength based on historical performance and relative skill level, we incorporated each team's Elo rating as a predictive feature in our model. This metric provides an objective strength metric that biases predictions toward higher-rated teams, reflecting their competitive advantage.

- **Teams** - A static reference table from the `nba_api` library containing essential team information, including full team names, standard abbreviations, and unique team identifiers. Since games in our Elo statistics dataset are only identified by team abbreviations and game date, this package was important in establishing proper joins across datasets and maintaining referential integrity across data sources.
- **LeagueDashPlayerStats** - A dataset, queried by date, from the `nba_api` library that provides cumulative season statistics for each player in the league up to the specified date. These statistics enable our model to account for the skill composition of each team's active lineup on the court, allowing for dynamic assessment of team strength based on which players are currently in play.
- **BoxScoreTraditionalV2** - This package provides complete lineup configurations and traditional box score statistics for each game of the NBA season. We utilize this dataset to initialize starting lineups and track lineup changes throughout each game. Since play descriptions in the play-by-play dataset denote substitutions using only player last names, this dataset is essential for accurately identifying players and updating active lineups in real time. This will allow us to retrieve the corresponding players' cumulative season statistics.

B. Data Preprocessing and Cleaning

Our preprocessing step focused on cleaning and aligning the play-by-play logs, which tend to contain inconsistencies and missing values due to the rapid logging format used by sports data feeds. This also included combining team and player stats to the play-by-play logs to allow for a more robust array of features in modeling. This step is critical for the success of the modeling and simulation modules, as time-aligned and semantically consistent input is required for effective prediction.

The messiness of raw play-by-play data has proven to be more severe than expected. While NBA API is structured well, inconsistency in naming conventions, event overlaps, and inconsistent logging coverage have made it difficult to cleanly segment actions. We have had to spend significant time designing preprocessing scripts that correct or realign these logs.

IV. METHODOLOGY

A. Overall Framework

Our proposed solution follows this data science pipeline:

- 1) Data acquisition - Collecting pregame and live data via NBA APIs and Elo datasets
- 2) Data preprocessing - Cleaning and aligning game events by time
- 3) Modeling - Training machine learning models (regressors, gradient boosting, neural networks, etc) on historical data in order to predict various outcome probabilities
- 4) LLM - Alongside predictive modeling, we will incorporate large language models through prompt engineering to automatically generate contextual insights. These

models will translate raw probabilities and what-if scenarios into natural-language summaries, enabling users to understand not just the numbers but the reasoning behind them.

- 5) Integration of our novel aspect - Create and embed the what-if simulator for user interaction
- 6) Visualization - Development of the dashboard to show live odds and scenario outcomes
- 7) Deployment - Host on an advanced cluster in order to handle the stream of live data

We leverage Python libraries such as pandas for data pre-processing, and scikit-learn to develop baseline models. Data storage and synchronization will be managed through structured databases to ensure scalability. Evaluation of early models will be benchmarked against naive baselines, such as assuming the team currently leading will win, to demonstrate clear improvement.

B. Feature Engineering

Feature engineering was a critical component of our modeling pipeline, transforming raw play-by-play data and various statistical sources into a coherent set of predictive features. Our feature engineering process focused on capturing three key aspects of game dynamics: current game state, team strength indicators, and player performance metrics.

1) *Game State Features*: The most fundamental feature we engineered was the live score margin, calculated Home Team Score - Away Team Score at each play-by-play event. This feature serves as a direct indicator of the current game state and provides the model with real-time context about which team is leading and by how much. We split the original combined score column into separate Home and Away score features to enable this calculation and provide the model with granular scoring information throughout the game. The score margin feature captures momentum shifts and allows the model to weight the current game state heavily in its probability estimates.

2) *Team Performance Aggregates*: To incorporate team-level performance capabilities, we engineered dynamic lineup-based aggregates that reflect the current on-court output at each moment of the game. Using the BoxScoreTraditionalV2 dataset to track substitutions and maintain real-time lineup information, we identified the five active players for each team at every play-by-play event. For these active players, we retrieved their cumulative season statistics from the LeagueDashPlayerStats dataset up to the game date, including points per game (PPG), assists per game (APG), rebounds per game (RPG), and plus-minus differential. We then summed these individual player statistics to create team-level aggregates that represent the collective performance capabilities of the players currently on the floor. This approach enables direct comparison between the Home Team's on-court lineup statistics and the Away Team's lineup statistics at any given moment. By calculating these metrics cumulatively up to the date of each game for our model training, we ensure that our model has access to the most current performance trends for a given game without introducing data leakage from future games. These dynamic

aggregate features enable the model to distinguish between lineups with consistently strong offensive output versus those relying more on defensive play or rebounding, and critically, to account for the significant impact of substitutions when star players enter or exit the game, providing context for interpreting in-game events and probability shifts.

3) *Team Strength Indicators*: We incorporated Elo ratings as a pregame strength indicator for both teams, engineering the Elo difference as $Elo(\text{Home Team}) - Elo(\text{Away Team})$. This single feature encapsulates historical performance and relative team quality in a probabilistic framework, similar to the approach used by Inpredictable [6]. The Elo difference serves as a prior probability that biases predictions toward higher-rated teams, while still allowing in-game events to override this prior as the game progresses. This feature proved particularly valuable in accounting for mismatches between teams of significantly different skill levels.

4) *Target Variable*: Our target variable for model training is a binary indicator of Home Team victory, encoded as 1 if the Home Team wins and 0 if the Home Team loses. This binary classification framework aligns with standard win probability modeling approaches and enables the use of logistic regression and similar probabilistic classifiers that output values between 0 and 1, which can be directly interpreted as win probabilities.

C. Modeling Approaches

Our initial modeling approach focused on logistic regression as a baseline model for predicting Home Team win probability at each game state. Logistic regression was selected for its interpretability, computational efficiency, and natural probabilistic output, which can be directly interpreted as win probability values between 0 and 1. The model predicts the probability that the Home Team wins the game given the current game state features. This formulation naturally extends to Away Team win probability through the complementary probability relationship, where $P(\text{Away Team Win}) = 1 - P(\text{Home Team Win})$.

Given the substantial size of our dataset, approximately 1,000,000 rows of play-by-play observations spanning multiple NBA seasons, we leveraged the Bridges-2 supercomputer infrastructure at the Pittsburgh Supercomputing Center for model training. The computational resources provided by Bridges-2 were essential for efficiently processing this large-scale dataset and enabled the quick iteration of features and model hyperparameters for our model.

V. WEEK 6 PROGRESS

A. Completed Tasks

By Week 6, we focused on establishing the foundational data infrastructure necessary for model development. We successfully set up a real-time data ingestion pipeline using the NBA API Python package, focusing initially on the NBA domain for proof of concept. The following endpoints were integrated:

- Play-by-play event logs: Sequential time-stamped records of every action in a game.

- Hustle stats: Advanced metrics such as deflections, contested shots, and box-outs.
- League standings and game context: To add metadata to each simulation.

We completed the initial data preprocessing pipeline, which involved cleaning and aligning play-by-play logs and combining team and player statistics with the play-by-play data to create a comprehensive feature set. This preprocessing work laid the groundwork for feature engineering that would be completed in subsequent weeks.

VI. WEEK 9 PROGRESS

A. Completed Tasks Since Week 6

Since Week 6, we made substantial progress in both dataset construction and model development. We successfully completed the feature engineering pipeline which included, transforming the raw play-by-play data into a structured dataset with 19 engineered features capturing game state, lineup-based performance metrics, and team strength indicators. The final dataset contains approximately 1,000,000 play-by-play observations spanning multiple NBA seasons, with each row representing a distinct game state and its associated features.

We developed Python scripts optimized for execution on the Bridges-2 supercomputer, enabling efficient model training on this large-scale dataset. These scripts include data loading routines, feature preprocessing pipelines, model training procedures, and comprehensive evaluation metrics. The design of these scripts facilitates rapid experimentation with different model architectures and hyperparameters.

B. Model Performance and Results

Our initial logistic regression model achieved strong performance on the win probability prediction task. Training on 643,640 observations and evaluating on a held-out test set of 160,911 observations, the model achieved a test accuracy of 79.15%, with precision of 80.44% and recall of 84.62%, resulting in an F1-score of 82.47%. The ROC-AUC score of 87.41% indicates strong discriminative ability in separating winning and losing game states.

The consistency between training metrics (79.23% accuracy, 87.44% ROC-AUC) and test metrics suggests that the model generalizes well without significant overfitting, despite the large feature space. This generalization performance validates our feature engineering approach and confirms that the selected features capture meaningful patterns in game dynamics.

Analysis of feature importance revealed that score margin is by far the most influential predictor, with a coefficient of 1.54, confirming the intuitive understanding that the current point differential is the strongest indicator of win probability. Visitor plus-minus total (0.58) and plus-minus differential (-0.58) emerged as the second and third most important features, highlighting the predictive value of lineup quality metrics. The Elo difference feature showed moderate importance (0.28), indicating that pregame team strength provides meaningful context but is appropriately downweighted relative to in-game performance. Game progress and seconds remaining

showed symmetric importance (0.15 and -0.15), capturing the temporal dynamics of win probability as games approach their conclusion.

Interestingly, some features showed counterintuitive coefficient signs, such as Home plus-minus total having a negative coefficient (-0.36). This likely reflects multicollinearity between related features and suggests opportunities for feature refinement in future iterations.

C. Dashboard Development

We initiated development of the interactive dashboard interface that will serve as the primary user-facing component of our system.

[Details about dashboard framework, visualization libraries, and initial design to be added]

VII. IMPLEMENTATION PLAN FOR FINAL DELIVERABLE

VIII. CONCLUSION

By the end of the project, we will deliver a scalable, real-time sports analytics dashboard with both predictive and interactive components. This system not only enhances fan engagement but also contributes a novel feature to sports data science: a dynamic simulator that merges live analytics with hypothetical exploration. Looking ahead, we envision our system not only serving as a fan engagement tool but also as a foundation for advanced decision support systems in sports analytics. By bridging live predictive modeling with interactive simulation, this project sets the stage for future innovations in real-time strategy optimization, broadcast enhancements, and even sports betting integrity analysis. The integration of LLMs ensures that our dashboard communicates results in a human-readable way, bridging the gap between complex predictive analytics and fan engagement. This capability positions the project as not only a quantitative tool but also a narrative platform for real-time sports analysis.

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REFERENCES

- [1] T. Horvat and J. Job, "The use of machine learning in sport outcome prediction: A review," *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, vol. 10, no. 5, pp. e1380, 2020.
- [2] ESPN, "Basketball Power Index," *ESPN NBA Statistics*, <https://www.espn.com/nba/bpi>.
- [3] FiveThirtyEight, "NBA Predictions," *FiveThirtyEight Projects*, <https://projects.fivethirtyeight.com/2025-nba-predictions/>.
- [4] R. Bunker and T. Susnjak, "The application of machine learning techniques for predicting match results in team sport: A review," *Journal of Artificial Intelligence Research*, vol. 73, pp. 1285–1322, 2022.

- [5] P. Robberechts, J. Van Haaren, and J. Davis, “Who will win it? An in-game win probability model for football,” *Machine Learning and Data Mining for Sports Analytics (ECML/PKDD 2019 Workshop)*, vol. abs/1906.05029, pp. 1–13, 2019.
- [6] M. Beuoy, “Updated NBA win probability calculator,” *Inpredictable*, Feb. 2015. [Online]. <https://www.inpredictable.com/2015/02/updated-nba-win-probability-calculator.html>
- [7] R. P. Bunker and F. Thabtah, “A machine learning framework for sport result prediction,” *Applied Computing and Informatics*, vol. 15, no. 1, pp. 27–33, 2019.